

For one view of Shannon's contributions, we have www.nyu.edu/pages/linguistics/courses/v610003/shan.html; for a biography, <http://scienceworld.wolfram.com/biography/Shannon.html>; for an excellent overview of Shannon's work, www.ams.org/notices/200201/fea-shannon.pdf.

And as if all that weren't enough for one decade, we have...

1.9 Atanasoff And Zuse

1939

Dr John Vincent Atanasoff and graduate student Clifford Berry of Iowa State College (now Iowa State University) designed a prototype 16-bit adder. This was the first machine to *calculate* using vacuum tubes.

Summer Of 1941

Atanasoff and Berry then completed a special-purpose calculator for solving simultaneous linear equations; this was later called the ABC (Atanasoff-Berry Computer). The clock speed was 60 Hz, and an addition took 1 second. For secondary memory—"hard disks", as we know them today—it used punch cards. The holes were not punched into the cards but burnt. For information on the ABC—if you can deal with a page that pitches Atanasoff as the Inventor of the "Real Computer"—there's www.intercom.net/local/shore_journal/joc10225.html.

December 1941

With some backing from the Deutsche Versuchsanstalt für Luftfahrt (German Aeronautical Research Institute), Konrad Zuse, a name you cannot afford to forget, completed his V3 (later the Z3): *the first operational programmable calculator*. So what? Well, computermuseum.li states:

"Dr Zuse's Z3 Computer, designed and built from 1938 to 1941, was the first automatic, program-controlled, fully-functional, general purpose digital computer."